

Hafiz Muhammad Awais

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SUMMARY

AI Engineer and Data Scientist with 3+ years of experience delivering end-to-end machine learning systems spanning computer vision, NLP, and agentic AI pipelines. IEEE-published researcher (2025) with live production deployments on AWS Lambda and HuggingFace Spaces. Instructor at FAST-NUCES delivering ML and data science labs. Proficient in PyTorch, LangGraph, FastAPI, and Docker. Holder of 39 verified certifications from IBM, DeepLearning.AI, Stanford, and Google.

TECHNICAL SKILLS

Languages:	Python, TypeScript, SQL
ML / Deep Learning:	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Optuna, SHAP
Computer Vision:	EfficientNetV2, ResNet, U-Net, Vision Transformer (ViT), OpenCV
NLP & Generative AI:	LangGraph, LangChain, RAG, HuggingFace Transformers, DistilBERT, Groq, Google Gemini, OpenAI API, Chainlit, NLTK
Backend / APIs:	FastAPI, Flask, Uvicorn, Gunicorn, REST APIs, Server-Sent Events
Frontend:	React 19, TypeScript, Vite, Tailwind CSS, Zustand
Cloud & DevOps:	Docker, Docker Compose, AWS Lambda, AWS S3, AWS ECR, HuggingFace Spaces
Data & Visualization:	Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel

EXPERIENCE

Instructor — ML & Programming Labs · FAST-NUCES, Islamabad Jan 2025 – Present

- Deliver weekly lab sessions for *Programming for Finance* and *Machine Learning for Business Analytics* using Python and Jupyter Notebooks for undergraduate cohorts.
- Design progressive project-based exercises on real-world financial datasets covering classification, customer churn prediction, and NLP pipelines.
- Conduct live coding sessions, mentor students on model debugging and evaluation, and administer graded lab assessments.

Freelance AI Engineer · Remote 2022 – Present

- Designed and deployed deep learning models (ResNet, U-Net, ViT) for medical image classification tasks, improving client accuracy from **85% to 95%** on held-out test sets.
- Engineered and shipped RetinaScan — a production medical imaging system on AWS Lambda using EfficientNetV2-S, FastAPI, and Docker for serverless real-time inference.
- Developed NLP tools including a financial news sentiment classifier (DistilBERT, **87.4% accuracy**) and a RAG-powered domain chatbot (LangChain, Chainlit).
- Delivered complete ML pipelines with real-time inference endpoints for healthcare and finance clients, including full feature engineering and model evaluation reporting.

PROJECTS

Selected projects listed below. Full portfolio available at github.com/hafiz-m-awais.

InsightForge AI · No Coding Multi-Agent ML Platform [\[Live\]](#) 2026 – Current

LangGraph · FastAPI · React 19 · TypeScript · Docker · XGBoost · LightGBM · Optuna · SHAP · Groq · Gemini · OpenAI

- Built a full-stack, no-code ML platform where users upload any dataset and receive a production-ready pipeline in minutes; 8 specialized LangGraph agents autonomously handle profiling, EDA, cleaning, feature engineering, model training, hyperparameter tuning, evaluation, and report generation across a 14-step workflow.
- Engineered a unified LLM router supporting Groq, Gemini, OpenAI, and OpenRouter; persisted feature engineering transforms in `.joblib` artifacts for correct production-time inference.
- Containerized with Docker Compose and deployed live on HuggingFace Spaces; React 19 + TypeScript frontend with Zustand state management served via a single FastAPI process.

RetinaScan — Diabetic Retinopathy Detection · IEEE Published [\[Live\]](#) 2026 – Current

PyTorch · EfficientNetV2-S · FastAPI · Docker · AWS Lambda · AWS ECR · AWS S3

- Fine-tuned EfficientNetV2-S on the APTOS 2019 fundus image dataset for 5-class DR severity grading; research published in IEEE (2025).

- Deployed serverlessly on AWS Lambda + ECR eliminating provisioning costs; drag-and-drop S3 frontend returns per-class confidence scores in real time.
- Implemented complete preprocessing pipeline including normalization, augmentation, and class-imbalance handling for robust clinical-grade inference.

NeuroTutor — Real-Time AI Tutoring Platform · [Live on HuggingFace Spaces](#) 2026
Flask · Google Gemini API · Server-Sent Events · Pyodide · Flask-Limiter · Docker · Gunicorn

- Built an AI tutor powered by Google Gemini with Server-Sent Events streaming; integrated in-browser Python execution via Pyodide so students run code without a backend dependency.
- Implemented abuse prevention via Flask-Limiter; containerized with Docker + Gunicorn and deployed on HuggingFace Spaces.

RAG Chatbot — Pakistan Knowledge Base · [Github](#) 2024
LangChain · Chainlit · PyTorch · HuggingFace Transformers

- Built a Retrieval-Augmented Generation chatbot grounded in a custom knowledge base covering Pakistani history, geography, economy, and culture; curated the dataset from scratch.
- Designed the full RAG pipeline: document embedding, vector retrieval (LangChain), LLM-grounded generation, and a real-time Chainlit GUI.

RESEARCH & PUBLICATIONS

- **Improving Inference Time in Diabetic Retinopathy — Recent Trends and Future Directions**
IEEE, 2025 ieeexplore.ieee.org/document/11153952
- **MS Thesis:** Optimized Feature Selection for Diabetic Retinopathy Detection using Deep Learning Models
FAST-NUCES, Islamabad

EDUCATION

Master of Science in Data Science · FAST-NUCES (National University of Computer & Emerging Sciences) Islamabad

- Thesis: *Optimized Feature Selection for Diabetic Retinopathy Detection using Deep Learning Models*
- IEEE publication: [Improving Inference Time in Diabetic Retinopathy — Recent Trends and Future Directions \(2025\)](#)

CERTIFICATIONS

- IBM Professional Data Science Certificate IBM / Coursera, 2024
- Machine Learning Specialization (3-course series) DeepLearning.AI / Andrew Ng, 2024
- Retrieval-Augmented Generation (RAG) DeepLearning.AI, 2025
- Applied Data Science Capstone IBM / Coursera, 2024

39 total verified certifications from IBM, DeepLearning.AI, Stanford University, Google, Coursera, and DataCamp.

PROFESSIONAL REFERENCES

Dr. Muhammad Nouman Noor — Assistant Professor, AI & Data Science
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Additional references available upon request.